

Dear Editor,

Thank you very much for arranging to review our manuscript. I have addressed all the criticisms/questions raised by the referee. Detailed responses to the queries of the referee have been provided below (Referee's comments are highlighted in blue). In view of these changes, I trust the manuscript will now be judged suitable for publication.

Sincerely,
Ipsita Mandal

REVIEWER 1

1. Points to improve in the introduction

- A long and complex sentence can make reading laborious. Rephrasing it into several shorter sentences would improve clarity. Make the reading more fluid through segmentation and reformulation.
- Emphasize a bit more at the beginning the physical interest or experimental relevance (to motivate less specialized readers).
- Include an opening sentence introducing the importance and potential applications of TSMs from the start.

I have updated everything according to the referee's suggestions. I have added the following sentences in appropriate places:

"TSMs provide a platform for us to formulate and observe some generic physical phenomena connected with quantum anomalies, especially in the context of multifold fermions. In the context of high-energy physics, chiral anomaly is synonymous with the Adler-Bell-Jackiw anomaly of quantum electrodynamics [1, 2]. Strikingly, this phenomenon continues to show up even in nonrelativistic settings involving 3d semimetals [3–5]. In particular, chiral anomaly from three-component quasiparticles is intriguing as they have no counterpart in relativistic spinors appearing in high-energy physics. This is because spin-statistics theorem prohibits the existence of fermions carrying an integer-spin value of unity."

"Their relevance in transport experiments involving multifold nodal points cannot be overemphasised, where the unmissable signatures of chiral anomaly have been observed via planar-Hall set-ups [6]. In particular, these experimental results and the associated theoretical explanations show that chiral anomaly is a generic phenomenon, not necessarily limited to spin-1/2 or pseudospin-1/2 fermions."

"The chirality is the origin of the chiral anomaly in topological nodal points."

Regarding long sentences, I do not quite know which ones the referee is referring to, as "long" is a qualitative description and what is "long" depends on individual preferences/tastes and writing style. If the referee feels there are some of those still remaining, it will be helpful if s/he explicitly points those out and I will try to modify them.

2. Could you more clearly specify the main originality of this article compared to existing works, especially those cited? What concrete impacts does this work have on the understanding or experimental investigation of pseudospin-1 semimetals?

We already had a few lines explaining these points in the original manuscript, viz.,

"For example, in our earlier works, we have computed the weak-magnetic-field-induced response in planar-Hall set-ups, in the limit of relaxation-time approximation (RTA). Here, we aim to perform an exact calculation of the longitudinal conductivity, arising from applying collinear electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields, by going beyond the RTA. This exercise has been carried out for WSMs [7, 8], which we will carry out for TSMs, whose unphysical representation of a flat-band is remedied by adding a spherically-symmetric quadratic-in-momentum correction [9]. Hence, the flat-band will be promoted to a quadratically-dispersive band, which will now contribute to the net conductivity (as expected in an actual experiment [6]). We would like to point out that, in Ref. [10], the behaviour of the longitudinal conductivity has been discussed for TSMs with a nondispersive band, where it does not contribute, since quadratic corrections are not taken into account."

Nevertheless, we have added some more explanations as shown below:

"To summarise the importance of our work, we emphasise on the following points: This is the first paper to incorporate a proper quadratic correction in the so-called flat-band of the TSMs while computing conductivity. This situation is highly nontrivial because the flat-band possesses a nontrivial OMM, despite having vanishing BC. In all prior literature [10–13], the conductivity from the flat-band has been omitted since, at leading order, it has zero dispersion. But this omission is not physical, as explained earlier. As such, our analysis here represents a significant advancement in the study of multifold semimetals in 3d. Secondly, the technique of solving the Boltzmann's equations exactly (i.e., by going beyond the RTA) [7], when more than one band is participating in transport at each chiral node, has not been implemented earlier. Our study of TSMs allows exactly that, demonstrating how intraband and interband scattering amplitudes can tune inear response. Last but not the least, it is a nontrivial task to extract the analytical expressions even within the RTA. We do that as well and bring out the inadequacies in the results obtained from RTA. Our investigations are very timely, thanks to the ongoing experimental efforts to probe novel transport properties of multifold fermions [6]."

3. What are the precise limitations of your isotropic effective model with quadratic corrections? How does this model compare to band structures obtained from ab initio calculations for real materials?

The model is taken from ab initio calculations for real materials, viz., [9, 14, 15] and there are no limitations as such. We have added the following discussions:

“The model follows from the low-energy $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near a band-crossing point (taken to be the origin or $\mathbf{k} = 0$), yielding the Hamiltonian consistent with the symmetries of the little group near the degeneracy point. Refs. [14, 15] expand upto linear order in momentum and the resulting model is an isotropic Hamiltonian with a flat band. The detailed group-theory analysis of the quadratic-corrected model can be found in the “Supplementary Information” of Ref. [9], where the authors expand about the nodal point upto quadratic order in momentum (they have considered the material CoSi for their experiments). For these quadratic terms, anisotropy is possible [9], but we retain only the isotropic contributions.”

4. For what physical reason is the quadratic correction essential, if not merely to correct the flat band? Are there concrete experimental evidences confirming its importance in real TSMs?

We already wrote in the manuscript that “However, the flatness of this band is an artifact of retaining only the linear-in-momentum terms in the effective Hamiltonian in the vicinity of a given node ...”. Any continuum model is obtained from the low-energy $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near a band-crossing point (taken to be the origin or $\mathbf{k} = 0$), yielding the Hamiltonian consistent with the symmetries of the little group near the degeneracy point. This effective Hamiltonian is valid only in the vicinity of a node, i.e., at low energies compared to the location of the nodal point. If we go away from a nodal point, of course the bands fold eventually and the global structure of the bands are seen in bandstructures obtained via DFT or ab initio calculations. This is a well-understood fact. Two examples are shown in Fig. 1. Concrete

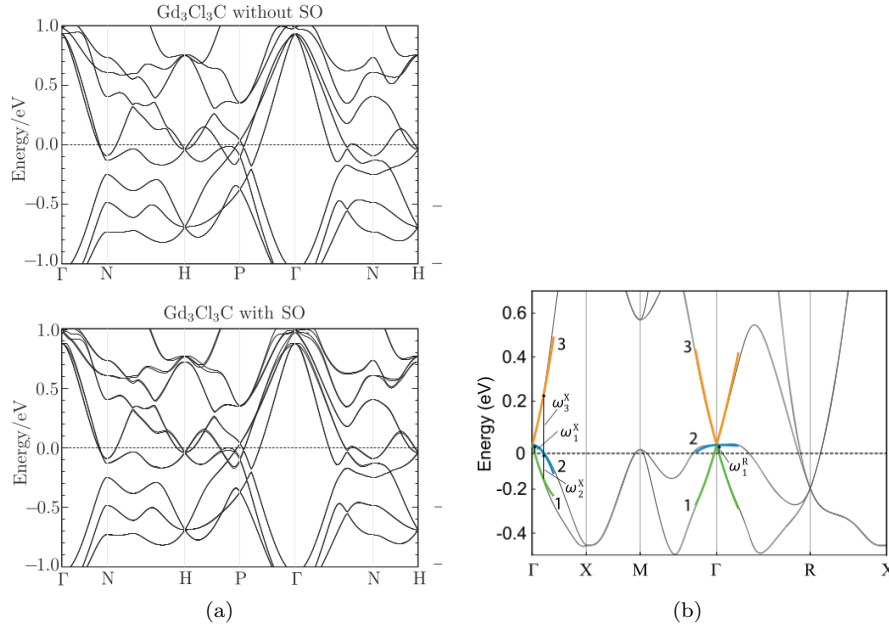


FIG. 1. (a) Band-structure $\text{Gd}_3\text{Cl}_3\text{C}$ without and with spin-orbit (SO) coupling, featuring a threefold crossing at the H point close to the Fermi level. Taken from Ref. [15]. (b) The band-structure of CoSi without SO coupling. A threefold node is seen at the Γ point. Taken from Ref. [9].

examples: Refs. [14, 15] expand upto linear order in momentum and the resulting model is an isotropic Hamiltonian with a flat band. The detailed group-theory analysis of the quadratic-corrected model can be found in the “Supplementary Information” of Ref. [9], where they expand about the nodal point upto quadratic order in momentum.

5. The description of intra- and interband scattering coefficients remains partly parametric. Do you have estimates or physical arguments for realistic bounds of these coefficients?

The absolute values are dependent on material properties and the concentration of impurities. Therefore, even for the same material, they are highly non-universal and impossible to assume without actually carrying out a measurement experiment. Hence, the logical way is to represent the dependence as ratios. On the other hand, the ratios of the scattering strengths for a given sample can be estimated by measuring the ratios of the relaxation times (see, for example, experiments carried out by Claudia Felser’s group [6]). This is the reason in the literature involving such theoretical studies [7, 8, 16], people have always considered ratios. What one could do for a given system is to set up scattering integrals in terms of Coulomb scattering by charged impurities. In the absence of screening, this of course gives a $1/q^2$ scattering potential. If the Fermi pockets are small, this situation can be approximated by one value

for intranode scattering and a smaller value for internode scattering. However, finite electron concentrations screen the potential in the usual way, giving $1/(q^2 + \eta^2)$. In the limit of strong screening, the ratio $\beta_{inter}/\beta_{inter}$ approaches unity. Additionally, there might be microscopic scattering mechanisms in a given material whose dominant long-range scatterings favour $\beta_{inter}/\beta_{inter} > 1$.

6. **Could you clarify the robustness and convergence of the ansatz method used for ? Have you tested alternative or more general forms?**

The method is completely self-consistent and what independent sinusoidal terms are to be included in $\Lambda_{\chi,s}^z$ are determined by looking at the form of $\mathcal{T}_{s,\tilde{s}}^{\chi,\tilde{\chi}}(\theta, \theta')$ [cf. Eq. (23)]. In the original version of the manuscript itself, we had pointed this out explicitly by the following statements:

“To solve the above equation, we take the self-consistent ansatz [7] that $\Lambda_{\chi,s}^z \equiv \Lambda_{\chi,s}^z(\mu, \theta)$ at an energy μ , which only depends on the polar angle, θ , and the chemical potential, μ . This is because, for elastic scattering, the integral over the full momentum space can be replaced by an integral over the Fermi surface at energy $\xi_{\chi,s}(\mathbf{k}) = \mu$, and the dependence on the azimuthal angle ϕ should not be there because of the rotational symmetry of the entire system in the $k_x k_y$ -plane. Consequently, the momentum-space integrals reduce to the respective Fermi surfaces at energy $\xi_{\chi,s}(k_F^{\chi,s}, \theta) = \mu$ with T set to zero, denoted by the set of Fermi momenta, $\{k_F^{\chi,s}(\theta)\}$. The self-consistency can be easily checked from the fact that $[\mathbf{\Omega}_s^{\chi}(\mathbf{k}) \cdot \mathbf{w}_{\chi,s}(\mathbf{k})]$ is ϕ -independent and $[\mathbf{w}_{\chi,s}(\mathbf{k}) \times \hat{\mathbf{z}}] \cdot \nabla_{\mathbf{k}} \Lambda_{\chi,s}^z(\mu, \theta)$ evaluates to zero.”

and

“Noting the form of the wavevectors and their overlaps [cf. Eqs. (22) and (23)], we now make the following ansatz:

$$\begin{aligned} \Lambda_{\chi,1}^z(\mu, \theta) &= \tau_{\chi,1}(\mu, \theta) \left[-h_{\chi,1}(\mu, \theta) + a_{\chi,1} \cos^4\left(\frac{\theta}{2}\right) + b_{\chi,1} \sin^4\left(\frac{\theta}{2}\right) + c_{\chi,1} \sin^2 \theta \right], \\ \Lambda_{\chi,0}^z(\mu, \theta) &= \tau_{\chi,0}(\mu, \theta) \left[-h_{\chi,0}(\mu, \theta) + c_{\chi,0} \sin^2 \theta + d_{\chi,0} \cos^2 \theta \right], \end{aligned} \quad (1)$$

”

Basically, one must ensure that $\Lambda_{\chi,s}^z(\mu, \theta) \Lambda_{\tilde{\chi},\tilde{s}}^z(\mu, \theta')$ contains all sinusoidal terms appearing in

$$\begin{aligned} \mathcal{T}_{s,\tilde{s}}^{\chi,\tilde{\chi}}(\theta, \theta') &= \left[\sin^4\left(\frac{\theta}{2}\right) \sin^4\left(\frac{\theta'}{2}\right) + \frac{1}{4} \sin^2 \theta \sin^2 \theta' + \cos^4\left(\frac{\theta}{2}\right) \cos^4\left(\frac{\theta'}{2}\right) \right] \delta_{s,1} \delta_{\tilde{s},1} \\ &+ \left[\frac{\sin^2 \theta \sin^2 \theta'}{2} + \cos^2 \theta \cos^2 \theta' \right] \delta_{s,0} \delta_{\tilde{s},0} \\ &+ \left[\left\{ \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right) \right\} \sin^2 \theta' + \sin^2 \theta \cos^2 \theta' \right] \delta_{s,1} \delta_{\tilde{s},0} \\ &+ \left[\sin^2 \theta \left\{ \sin^4\left(\frac{\theta'}{2}\right) + \cos^4\left(\frac{\theta'}{2}\right) \right\} + \cos^2 \theta \sin^2 \theta' \right] \delta_{s,0} \delta_{\tilde{s},1}, \end{aligned}$$

In fact, if you take more terms with independent sinusoidal forms, the solutions for the corresponding coefficients are zero. For example, in Ref. [10], the authors have included one extra spurious/unnecessary coefficient, λ^{χ} (accompanying unity or θ -independent term), which identically turns out to be zero [because $\mathcal{T}_{s,\tilde{s}}^{\chi,\tilde{\chi}}(\theta, \theta')$ does not contain any such θ - or θ' -independent term]. This has also been tested. The formalism is mathematically water-tight as it is self-consistent, and no other term can appear at least in the $\mathbf{E} \parallel \mathbf{B}$ case. In other words, there are NO ALTERNATIVE forms and the choice is unique. The methodology follows directly from Ref. [7].

Just to demonstrate how the ansatz changes depending on the system one considers, we provide the following examples:

- (a) For a pair of chirally-conjugate Weyl nodes (each with a twofold band-crossing) analysed in Ref. [7], the overlap-function is

$$\mathcal{T}_{\chi,\tilde{\chi}}(\theta, \theta') = \frac{1 + \chi \tilde{\chi} \cos \theta \cos \theta'}{2}.$$

Hence, the ansatz is taken to be

$$\Lambda_{\chi}^z(\mu, \theta) = \tau_{\chi}(\mu, \theta) [\lambda_{\chi} - h_{\chi}(\mu, \theta) + a_{\chi} \cos \theta],$$

- (b) For a single Kramers-Weyl node with a twofold band-crossing analysed in Ref. [17], the overlap-function is

$$\mathcal{T}_{s,\tilde{s}}(\theta, \theta') = \frac{1 + s \tilde{s} \cos \theta \cos \theta'}{2},$$

for which the ansatz is taken to be

$$\Lambda_s^z(\mu, \theta) = \tau_s(\mu, \theta) [\lambda_s - h_s(\mu, \theta) + a_s \cos \theta],$$

(c) For a single RSW node with a fourfold band-crossing analysed in Ref. [18], the overlap-function is

$$\begin{aligned} \mathcal{T}_{s,\tilde{s}}(\theta, \theta') = & [5 - 3 \cos^2 \theta' + \cos \theta (17 \cos \theta' - 27 \cos^3 \theta') + \cos^2 \theta (9 \cos^2 \theta' - 3) + 9 \cos^3 \theta \cos \theta' (5 \cos^2 \theta' - 3)] \beta_{\text{intra}}^{1,1} \\ & + [5 - 3 \cos^2 \theta' + \cos \theta (9 \cos \theta' - 3 \cos^3 \theta') + \cos^2 \theta (9 \cos^2 \theta' - 3) + \cos^3 \theta (5 \cos^3 \theta' - 3 \cos \theta')] \beta_{\text{intra}}^{3,3} \\ & + [3 + 3 \cos^2 \theta' + \cos^3 \theta (9 \cos \theta' - 15 \cos^3 \theta') + \cos \theta (9 \cos^3 \theta' - 3 \cos \theta') + \cos^2 \theta (3 - 9 \cos^2 \theta')] \beta_{\text{inter}}. \end{aligned}$$

Accordingly, the ansatz is taken to be

$$\Lambda_s^z(\mu, \theta) = \tau_s(\mu, \theta) [\lambda_s - h_s + a_s \cos \theta + b_s \cos^2 \theta + c_s \cos^3 \theta].$$

For the benefit of the reader, these discussions are now added in an appendix.

7. What are the expected effects if spherical symmetry is broken, for example by anisotropy in the material? Could this case substantially modify your conclusions?

The ansatz will now have to be modified and will contain ϕ -dependent terms. This makes the numerics more difficult as we cannot perform the ϕ -integrals from the start, especially with the added complexity of multiple bands as well as an extra quadratic term in the dispersion.

8. You show that the $s = 0$ band, previously ignored, contributes significantly via the orbital magnetic moment (OMM). How does this contribution evolve with changes in doping or chemical potential in a real material?

We have already shown our results for some chosen chemical potential (μ). A positive chemical potential will cut the $s = 1$ and $s = 0$ bands, which is the case considered here. The OMM is not affected by μ — the Fermi momentum is determined by μ [see the discussions below Eq. (22)]. It is just that if OMM were zero, the conductivity from the $s = 0$ band would have vanished. The qualitative features will be the same if we vary μ within the low-energy range where the continuum model is valid. Of course, for a negative value of chemical potential will cut only the $s = -1$ band. Adding more plots for more values of μ will clutter the manuscript and will not lead to any new knowledge.

9. The gap between the exact solution and the relaxation-time approximation is highlighted. Can you quantify this “loss” in the approximation, and specify under which experimental conditions this would be observable?

Any experimental result should be interpreted using the exact solution, and NOT the relaxation-time approximation. The main message is that a theoretical modelling using the naive relaxation-time approximation will lead to wrong conclusions. Under ALL experimental conditions, the exact-solution results should be the one which must be considered. Our results highlight the failure of accuracy on resorting to naive relaxation-time approximations. People working with collision integrals in other contexts have also discovered RTA fails to yield correct results (see, for example Refs. [19, 20]) and one should not blindly count on RTA formalism. I have added the following lines:

“Indeed, the failure of an RTA has been observed in diverse contexts (see, for example Refs. [19, 20]), where the correct physics is only captured by an exact treatment of the collision integrals.”

10. The paper indicates an asymmetry between contributions from nodes with opposite chirality. Could this asymmetry lead to specific observable effects, for example in non-reciprocal transport?

The net electric current cannot distinguish it, as it is not. However, one may consider strain-induced pseudomagnetic field, for which the current will now be chirality-dependent [8, 21, 22]. This we defer to a future work.

11. Some figures show complex behaviors of conductivity as a function of ratios. Can you provide a more detailed physical explanation for the sign reversals observed?

We have added the following detailed discussions:

“Let us now dissect the origins of various contributions to the net magnetoconductance, trying to pin-point its sign upon including the OMM:

- Case I — The features of δ_{zz} must be the same as in the WSM case when we consider only the $s = \tilde{s} = 1$ bands, exemplified by Fig. 2(c). Let us label the BC-only, OMM-only, and the concurrent part of $\delta\sigma_{zz}$ as $\delta\sigma_{zz}^{BC}$, $\delta\sigma_{zz}^{OM}$, and $\delta\sigma_{zz}^{conc}$, respectively. The last one implies the part when both the BC and OMM contribute together. Indeed, we find that the sign change of $\delta\sigma_{zz}$ is not the result of a simple competition between $\delta\sigma_{zz}^{BC}$, $\delta\sigma_{zz}^{OM}$, as each of these two mechanisms individually produces a positive value. Only $\delta\sigma_{zz}^{conc}$ comes with a negative sign. This is because of the following: First we observe that $\eta_{\chi,bc} \equiv \text{sgn}[\mathbf{\Omega}_{\chi,1} \cdot \mathbf{B}]$ and $\eta_{\chi,m} \equiv \text{sgn}[\varepsilon_{\chi,1}^{(m)}]$ have opposite signs. Since the response always goes as B^2 , the surviving individual contributions appear as $\eta_{\chi,bc} \eta_{\tilde{\chi},bc}$ and $\eta_{\chi,m} \eta_{\tilde{\chi},m}$, respectively, which has a sign opposite to that of $\eta_{\chi,bc} \eta_{\tilde{\chi},m}$ appearing in $\delta\sigma_{zz}^{conc}$. As we crank up $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}}$, $|\sigma_{zz}^{conc}|$ increases and $\delta\sigma_{zz}^{BC} + \delta\sigma_{zz}^{OM}$ decreases in magnitude, with the former overtaking the latter eventually. For example, we find the following dataset for $B = 60 \text{ eV}^2$:
 $\delta\sigma_{zz} = 0.0174257$, $\delta\sigma_{zz}^{BC} = 0.0413964$, $\delta\sigma_{zz}^{OM} = 0.0377579$, and $\delta\sigma_{zz}^{conc} = -0.0617285$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 0.5$;
 $\delta\sigma_{zz} = 0.00277429$, $\delta\sigma_{zz}^{BC} = 0.0257312$, $\delta\sigma_{zz}^{OM} = 0.0204378$, and $\delta\sigma_{zz}^{conc} = -0.0433948$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 1$;
 $\delta\sigma_{zz} = -0.00376071$, $\delta\sigma_{zz}^{BC} = 0.0186851$, $\delta\sigma_{zz}^{OM} = 0.0122026$, and $\delta\sigma_{zz}^{conc} = -0.0346485$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 2$.

- Case II — For the case when we consider only the subsystem of $s = \tilde{s} = 0$ bands, exemplified by the lowermost row of Fig. 3, we note that we have no BC contribution whatsoever. The overall $\delta\sigma_{zz}$ is always negative irrespective of the value of $\beta_{0,0}^{\text{inter}}/\beta_{0,0}^{\text{intra}}$. We also note that the magnitude of $\delta\sigma_{zz}$ is roughly of the order of 10 times smaller than that for Case I.
- Case III — When the interplay of all the bands are included, the qualitative features of Case I and Case II continue to persist when we look at the contributions coming from the two types of bands. By observing that the magnitude of $\delta\sigma_{zz}$ is order of magnitude higher for the $s = \tilde{s} = 1$ bands than that for the $s = \tilde{s} = 0$ bands, the net conductivity coming from all the bands (which is what will be measured in an experiment) will be dominated by the former.

We must also warn the reader that the solutions are found by solving the complicated integro-differential equation shown in Eq. (22). As such, the exact features and origins cannot be determined more precisely. How the complicated angular dependence, scattering strengths, quadratic-in-momentum terms, conservation of net electric charge, etc. feed into the equations to bring out the final features can only be obtained via the actual solutions.”

12. Certain symbols (e.g., quadratic correction coefficient, scattering coefficients) should be explicitly defined at the beginning of the paper to facilitate reading.

All symbols have been defined at the appropriate places where they are first defined. Hence, there is no hindrance to reading as the paper progresses step-by-step, introducing the physical quantities, defining the equations to be solved, and eventually finding the solutions. In some books, there is a chapter/section defining all symbols used in the book, but I am unfamiliar with declaring all relevant symbols at the beginning of a paper (this is not a book) where the physical quantities have not even be defined. I have not seen it done in any paper and hence have no clue how to do it in the conventions for writing a manuscript. Declaring symbols all of a sudden in the beginning, without explaining the physical quantities, does not seem very logical to me.

13. The outlook mentions extensions to non-collinear fields and tilted spectra. Do you already have preliminary ideas or results to share?

No, I do not have any preliminary result and it is beyond the scope of this work. But the treatment will be similar to Ref. [8], but now with the added complexity of more number of bands as well as an extra quadratic term in the dispersion. Regarding tilt, it is not allowed for TSMs, as follows from the sentences I added about the effective model derived from first principles, viz.

“The model follows from the low-energy $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near a band-crossing point (taken to be the origin or $\mathbf{k} = 0$), yielding the Hamiltonian consistent with the symmetries of the little group near the degeneracy point. Refs. [14, 15] expand upto linear order in momentum and the resulting model is an isotropic Hamiltonian with a flat band. The detailed group-theory analysis of the quadratic-corrected model can be found in the “Supplementary Information” of Ref. [9], where they expand about the nodal point upto quadratic order in momentum. For these quadratic terms, anisotropy is possible [9], but we retain only the isotropic contributions.”

14. Which emerging candidate materials could provide direct experimental validation of your predictions? Are there specific transport signatures to expect?

Yes, there are some experiments by Claudia Felser’s group [6], where they have considered CoSi. They have measured negative magnetoresistance in planar-Hall set-ups with non-collinear electromagnetic fields, but fitted their data with a hand-waving argument involving RTA.

15. The paper mentions several related effects (anomalous Hall, Magnus effect, etc.): to what extent could the model you propose be adapted to compute these other responses?

The Boltzmann equations for these particular effects have to be written down and then the solutions has to be formulated. The solutions will crucially depend on the exact forms of the collision integrals for those individual cases.

16. Finally, what is the scope of semiclassicality in this context, especially at low temperatures and weak magnetic fields that are not quantizing? Is there a notable impact of Landau quantization that you do not address here?

Yes of course — Landau Level quantization completely change the forms of the quasiparticle dispersion from continuum bands to discrete levels. An example can be found in our earlier work, Ref. [23], where the conductivity will show quantum-Hall plateaus. These are completely two different regimes. An estimate of the upper limit of values of magnetic field for which the semiclassical formalism is valid can be found in the Supplementary Material of Ref. [7], which we do not repeat in our manuscript.

REVIEWER 2

1. The manuscript emphasizes the exact solution of the Boltzmann equation beyond the relaxation- time approximation (RTA). To what extent is this a significant advancement over previous works that relied on the RTA for similar systems?

We already had a few lines explaining these points in the original manuscript, viz.,

“For example, in our earlier works, we have computed the weak-magnetic-field-induced response in planar-Hall set-ups, in the limit of relaxation-time approximation (RTA). Here, we aim to perform an exact calculation of the longitudinal conductivity, arising from applying collinear electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields, by going beyond the RTA. This exercise has been carried out for WSMs [7, 8], which we will carry out for TSMs, whose unphysical representation of a flat-band is remedied by adding a spherically-symmetric quadratic-in-momentum correction [9]. Hence, the flat-band will be promoted to a quadratically-dispersive band, which will now contribute to the net conductivity (as expected in an actual experiment [6]). We would like to point out that, in Ref. [10], the behaviour of the longitudinal conductivity has been discussed for TSMs with a nondispersive band, where it does not contribute, since quadratic corrections are not taken into account.”

Nevertheless, we have added some more explanations as shown below:

“To summarise the importance of our work, we emphasise on the following points: This is the first paper to incorporate a proper quadratic correction in the so-called flat-band of the TSMs while computing conductivity. This situation is highly nontrivial because the flat-band possesses a nontrivial OMM, despite having vanishing BC. In all prior literature [10–13], the conductivity from the flat-band has been omitted since, at leading order, it has zero dispersion. But this omission is not physical, as explained earlier. As such, our analysis here represents a significant advancement in the study of multifold semimetals in 3d. Secondly, the technique of solving the Boltzmann’s equations exactly (i.e., by going beyond the RTA) [7], when more than one band is participating in transport at each chiral node, has not been implemented earlier. Our study of TSMs allows exactly that, demonstrating how intraband and interband scattering amplitudes can tune inear response. Last but not the least, it is a nontrivial task to extract the analytical expressions even within the RTA. We do that as well and bring out the inadequacies in the results obtained from RTA. Our investigations are very timely, thanks to the ongoing experimental efforts to probe novel transport properties of multifold fermions [6].”

2. The isotropic model with a specific quadratic correction (ck^2) is used to remedy the unphysical flat band. Are there specific material systems where this form of the low-energy Hamiltonian is known to be a valid approximation?

The model is taken from ab initio calculations for real materials, viz., [9, 14, 15] and there are no limitations as such. We have added the following discussions:

“The model follows from the low-energy $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near a band-crossing point (taken to be the origin or $\mathbf{k} = 0$), yielding the Hamiltonian consistent with the symmetries of the little group near the degeneracy point. Refs. [14, 15] expand upto linear order in momentum and the resulting model is an isotropic Hamiltonian with a flat band. The detailed group-theory analysis of the quadratic-corrected model can be found in the “Supplementary Information” of Ref. [9], where the authors expand about the nodal point upto quadratic order in momentum (they have considered the material CoSi for their experiments). For these quadratic terms, anisotropy is possible [9], but we retain only the isotropic contributions.”

3. How might the conclusions be modified in a more realistic, anisotropic system?

The ansatz will now have to be modified and will contain ϕ -dependent terms. This makes the numerics more difficult as we cannot perform the ϕ -integrals from the start. The analysis will be similar to the tilted Weyl cases studied in Ref. [8], but now with the added complexity of more number of bands as well as an extra quadratic term in the dispersion. Without having done the calculations, I cannot predict what the quantitative modifications will be. But the main point is that there will be nonzero transverse components of conductivity. An anisotropic system is not necessarily more realistic as these parameters are material-dependent.

4. The scattering potentials β are introduced as phenomenological parameters. Is there a physical justification for the specific relationships and symmetries assumed between them (e.g., $\beta_{inter}/\beta_{inter} > 1$)?

I believe the statement “ $\beta_{inter}/\beta_{inter} > 1$ ” in the query is a typo. The absolute values are dependent on material properties and the concentration of impurities. Therefore, even for the same material, they are highly non-universal and not dependent on any symmetries. The ratios of the scattering strengths for a given sample can be estimated by measuring the ratios of the relaxation times (see, for example, experiments carried out by Claudia Felser’s group [6]). What one could do for a given system is to set up scattering integrals in terms of Coulomb scattering by charged impurities. In the absence of screening, this of course gives a $1/q^2$ scattering potential. If the Fermi pockets are small, this situation can be approximated by one value for intranode scattering and a smaller value for internode scattering. However, finite electron concentrations screen the potential in the usual way, giving $1/(q^2 + \eta^2)$. In the limit of strong screening, the ratio $\beta_{inter}/\beta_{inter}$ approaches unity. Additionally, there might be microscopic scattering mechanisms in a given material whose dominant long-range scatterings favour $\beta_{inter}/\beta_{inter} > 1$.

5. How sensitive are the main results (e.g., the sign flip of $\delta\sigma_{zz}$) to the specific choices of the β ratios?

For a wide range of changes, they remain the same, as can be seen from the results and from the trials I have performed but not shown such a huge amount of data in the manuscript. I have chosen to show a few representative ones which show the validity in broad ranges.

6. The ansatz for the vector mean-free path $\Lambda_{\chi,s}(\mu, \theta)$ in Eq. 27 appears to be quite specific. What is the rationale behind the chosen angular dependencies (e.g., $\cos^4(\theta/2)$, $\sin^2 \theta$)? Is the completeness of this ansatz guaranteed?

Yes, the completeness of this ansatz is guaranteed as it exhausts any term that can appear on either side of the linearised Boltzmann equation. If you take more terms with independent sinusoidal forms, the solutions for the corresponding coefficients are zero. This has been tested. The formalism is mathematically water-tight as it is self-consistent, and no other NON-ZERO term can appear at least in the $\mathbf{E} \parallel \mathbf{B}$ case. Basically, one must ensure that $\Lambda_{\chi,s}^z(\mu, \theta) \Lambda_{\tilde{\chi},\tilde{s}}^z(\mu, \theta')$ contains all sinusoidal terms appearing in

$$\begin{aligned} \mathcal{T}_{s,\tilde{s}}^{\chi,\tilde{\chi}}(\theta, \theta') = & \left[\sin^4\left(\frac{\theta}{2}\right) \sin^4\left(\frac{\theta'}{2}\right) + \frac{1}{4} \sin^2 \theta \sin^2 \theta' + \cos^4\left(\frac{\theta}{2}\right) \cos^4\left(\frac{\theta'}{2}\right) \right] \delta_{s,1} \delta_{\tilde{s},1} \\ & + \left[\frac{\sin^2 \theta \sin^2 \theta'}{2} + \cos^2 \theta \cos^2 \theta' \right] \delta_{s,0} \delta_{\tilde{s},0} \\ & + \left[\left\{ \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right) \right\} \sin^2 \theta' + \sin^2 \theta \cos^2 \theta' \right] \delta_{s,1} \delta_{\tilde{s},0} \\ & + \left[\sin^2 \theta \left\{ \sin^4\left(\frac{\theta'}{2}\right) + \cos^4\left(\frac{\theta'}{2}\right) \right\} + \cos^2 \theta \sin^2 \theta' \right] \delta_{s,0} \delta_{\tilde{s},1}, \end{aligned}$$

In fact, if you take more terms with independent sinusoidal forms, the solutions for the corresponding coefficients are zero. For example, in Ref. [10], the authors have included one extra spurious/unnecessary coefficient, λ^χ (accompanying unity or θ -independent term), which identically turns out to be zero [because $\mathcal{T}_{s,\tilde{s}}^{\chi,\tilde{\chi}}(\theta, \theta')$ does not contain any such θ - or θ' -independent term]. This has also been tested. The formalism is mathematically water-tight as it is self-consistent, and no other term can appear at least in the $\mathbf{E} \parallel \mathbf{B}$ case. In other words, there are NO ALTERNATIVE forms and the choice is unique. The methodology follows directly from Ref. [7].

Just to demonstrate how the ansatz changes depending on the system one considers, we provide the following examples:

- (a) For a pair of chirally-conjugate Weyl nodes (each with a twofold band-crossing) analysed in Ref. [7], the overlap-function is

$$\mathcal{T}_{\chi,\tilde{\chi}}(\theta, \theta') = \frac{1 + \chi \tilde{\chi} \cos \theta \cos \theta'}{2}.$$

Hence, the ansatz is taken to be

$$\Lambda_{\chi}^z(\mu, \theta) = \tau_{\chi}(\mu, \theta) [\lambda_{\chi} - h_{\chi}(\mu, \theta) + a_{\chi} \cos \theta],$$

- (b) For a single Kramers-Weyl node with a twofold band-crossing analysed in Ref. [17], the overlap-function is

$$\mathcal{T}_{s,\tilde{s}}(\theta, \theta') = \frac{1 + s \tilde{s} \cos \theta \cos \theta'}{2},$$

for which the ansatz is taken to be

$$\Lambda_s^z(\mu, \theta) = \tau_s(\mu, \theta) [\lambda_s - h_s(\mu, \theta) + a_s \cos \theta],$$

- (c) For a single RSW node with a fourfold band-crossing analysed in Ref. [18], the overlap-function is

$$\begin{aligned} \mathcal{T}_{s,\tilde{s}}(\theta, \theta') = & [5 - 3 \cos^2 \theta' + \cos \theta (17 \cos \theta' - 27 \cos^3 \theta') + \cos^2 \theta (9 \cos^2 \theta' - 3) + 9 \cos^3 \theta \cos \theta' (5 \cos^2 \theta' - 3)] \beta_{\text{intra}}^{1,1} \\ & + [5 - 3 \cos^2 \theta' + \cos \theta (9 \cos \theta' - 3 \cos^3 \theta') + \cos^2 \theta (9 \cos^2 \theta' - 3) + \cos^3 \theta (5 \cos^3 \theta' - 3 \cos \theta')] \beta_{\text{intra}}^{3,3} \\ & + [3 + 3 \cos^2 \theta' + \cos^3 \theta (9 \cos \theta' - 15 \cos^3 \theta') + \cos \theta (9 \cos^3 \theta' - 3 \cos \theta') + \cos^2 \theta (3 - 9 \cos^2 \theta')] \beta_{\text{inter}}. \end{aligned}$$

Accordingly, the ansatz is taken to be

$$\Lambda_s^z(\mu, \theta) = \tau_s(\mu, \theta) [\lambda_s - h_s + a_s \cos \theta + b_s \cos^2 \theta + c_s \cos^3 \theta].$$

For the benefit of the reader, these discussions are now added in an appendix.

7. The matrix equation in the Appendix is central to the solution. Are the matrix elements A and H provided with sufficient detail for the results to be reproducible?

Of course – there is no question about it. It is obtained by plugging in the ansatz in the Boltzmann equation. The long matrix equation has been written out explicitly in the appendix so that it is easy for the reader to perform this exercise and check if they got all the terms correctly.

8. The paper clearly shows that the RTA fails to capture the rich behavior (like sign flips and negative LMC) found in the exact solution. Does the analysis reveal why the RTA fails so fundamentally in this multifold system? Is it primarily due to the neglect of interband scattering or the specific treatment of the collision integral?

Yes, I have repeatedly emphasised that — it is primarily due to the specific treatment of the collision integral. People working with collision integrals in other contexts have also discovered RTA fails to yield correct results (see, for example Refs. [19, 20]) and one should not blindly count on RTA formalism. I have added the following lines:

“Indeed, the failure of an RTA has been observed in diverse contexts (see, for example Refs. [19, 20]), where the correct physics is only captured by an exact treatment of the collision integrals.”

9. A key result is the negative longitudinal magnetoconductivity (LMC) for the $s = 0$ band and under certain conditions for the $s = 1$ bands. What is the precise physical mechanism originating from the OMM and interband scattering that causes this negative LMC?

We have added the following detailed discussions:

“Let us now dissect the origins of various contributions to the net magnetoconductance, trying to pin-point its sign upon including the OMM:

- Case I — The features of $\delta\sigma_{zz}$ must be the same as in the WSM case when we consider only the $s = \tilde{s} = 1$ bands, exemplified by Fig. 2(c). Let us label the BC-only, OMM-only, and the concurrent part of $\delta\sigma_{zz}$ as $\delta\sigma_{zz}^{BC}$, $\delta\sigma_{zz}^{OM}$, and $\delta\sigma_{zz}^{conc}$, respectively. The last one implies the part when both the BC and OMM contribute together. Indeed, we find that the sign change of $\delta\sigma_{zz}$ is not the result of a simple competition between $\delta\sigma_{zz}^{BC}$, $\delta\sigma_{zz}^{OM}$, as each of these two mechanisms individually produces a positive value. Only $\delta\sigma_{zz}^{conc}$ comes with a negative sign. This is because of the following: First we observe that $\eta_{\chi,bc} \equiv \text{sgn}[\mathbf{\Omega}_{\chi,1} \cdot \mathbf{B}]$ and $\eta_{\chi,m} \equiv \text{sgn}[\varepsilon_{\chi,1}^{(m)}]$ have opposite signs. Since the response always goes as B^2 , the surviving individual contributions appear as $\eta_{\chi,bc} \eta_{\tilde{\chi},bc}$ and $\eta_{\chi m} \eta_{\tilde{\chi} m}$, respectively, which has a sign opposite to that of $\eta_{\chi,bc} \eta_{\tilde{\chi},m}$ appearing in $\delta\sigma_{zz}^{conc}$. As we crank up $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}}$, $|\sigma_{zz}^{conc}|$ increases and $\delta\sigma_{zz}^{BC} + \delta\sigma_{zz}^{OM}$ decreases in magnitude, with the former overtaking the latter eventually. For example, we find the following dataset for $B = 60$ eV²:
 $\delta\sigma_{zz} = 0.0174257$, $\delta\sigma_{zz}^{BC} = 0.0413964$, $\delta\sigma_{zz}^{OM} = 0.0377579$, and $\delta\sigma_{zz}^{conc} = -0.0617285$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 0.5$;
 $\delta\sigma_{zz} = 0.00277429$, $\delta\sigma_{zz}^{BC} = 0.0257312$, $\delta\sigma_{zz}^{OM} = 0.0204378$, and $\delta\sigma_{zz}^{conc} = -0.0433948$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 1$;
 $\delta\sigma_{zz} = -0.00376071$, $\delta\sigma_{zz}^{BC} = 0.0186851$, $\delta\sigma_{zz}^{OM} = 0.0122026$, and $\delta\sigma_{zz}^{conc} = -0.0346485$ for $\beta_{1,1}^{\text{inter}}/\beta_{1,1}^{\text{intra}} = 2$.
- Case II — For the case when we consider only the subsystem of $s = \tilde{s} = 0$ bands, exemplified by the lowermost row of Fig. 3, we note that we have no BC contribution whatsoever. The overall $\delta\sigma_{zz}$ is always negative irrespective of the value of $\beta_{0,0}^{\text{inter}}/\beta_{0,0}^{\text{intra}}$. We also note that the magnitude of $\delta\sigma_{zz}$ is roughly of the order of 10 times smaller than that for Case I.
- Case III — When the interplay of all the bands are included, the qualitative features of Case I and Case II continue to persist when we look at the contributions coming from the two types of bands. By observing that the magnitude of $\delta\sigma_{zz}$ is order of magnitude higher for the $s = \tilde{s} = 1$ bands than that for the $s = \tilde{s} = 0$ bands, the net conductivity coming from all the bands (which is what will be measured in an experiment) will be dominated by the former.

We must also warn the reader that the solutions are found by solving the complicated integro-differential equation shown in Eq. (22). As such, the exact features and origins cannot be determined more precisely. How the complicated angular dependence, scattering strengths, quadratic-in-momentum terms, conservation of net electric charge, etc. feed into the equations to bring out the final features can only be obtained via the actual solutions.”

10. Certain symbols (e.g., quadratic correction coefficient, scattering coefficients) should be explicitly defined at the beginning of the paper to facilitate reading.

All symbols have been defined at the appropriate places where they are first defined. Hence, there is no hindrance to reading as the paper progresses step-by-step, introducing the physical quantities, defining the equations to be solved, and eventually finding the solutions. In some books, there is a chapter/section defining all symbols used in the book, but I am unfamiliar with declaring all relevant symbols at the beginning of a paper (this is not a book) where the physical quantities have not even be defined. I have not seen it done in any paper and hence have no clue how to do it in the conventions for writing a manuscript. Declaring symbols all of a sudden in the beginning, without explaining the physical quantities, does not seem very logical to me.

11. The outlook mentions extensions to non-collinear fields and tilted spectra. Do you already have preliminary ideas or results to share?

No, I do not have any preliminary result and it is beyond the scope of this work. But the treatment will be similar to Ref. [8], but now with the added complexity of more number of bands as well as an extra quadratic term in the dispersion. Regarding tilt, it is not allowed for TSMs, as follows from the sentences I added about the effective model derived from first principles, viz.

“The model follows from the low-energy $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near a band-crossing point (taken to be the origin or $\mathbf{k} = 0$),

yielding the Hamiltonian consistent with the symmetries of the little group near the degeneracy point. Refs. [14, 15] expand upto linear order in momentum and the resulting model is an isotropic Hamiltonian with a flat band. The detailed group-theory analysis of the quadratic-corrected model can be found in the “Supplementary Information” of Ref. [9], where they expand about the nodal point upto quadratic order in momentum. For these quadratic terms, anisotropy is possible [9], but we retain only the isotropic contributions.”

12. The results show an asymmetry between the $\chi = +1$ and $\chi = -1$ nodes for the $s = 0$ band. What is the origin of this chirality-dependent response, given that the BC is zero for this band?

It has already been emphasised in the manuscript that, although the BC is zero for the $s = 0$ band, its OMM is non-zero which endows it with nontrivial topological properties. See Eq. (8) and the lines below it:

“We note that, for the $s = 0$ band, although the BC is identically zero. However, the OMM is nonzero, turning out to be twice the OMM of the $s = \pm 1$ bands. Therefore, all the bands are endowed with nontrivial topological properties. It is easy to verify from the $\Omega_{\chi,s}(\mathbf{k})$ -expressions that the node has a net Chern number of 2χ , contributed by the $s = \pm 1$ bands.”

The sign of the OMM is χ -dependent.

13. In the Figures (e.g., 2, 3, 4), what are the absolute values of the scattering parameters β used? Providing only ratios might not be sufficient to gauge the overall scattering strength.

The absolute values are dependent on material properties and the concentration of impurities. Therefore, even for the same material, they are highly non-universal and impossible to assume without actually carrying out a measurement experiment. Hence, the logical way is to represent the dependence as ratios. On the other hand, the ratios of the scattering strengths for a given sample can be estimated by measuring the ratios of the relaxation times (see, for example, experiments carried out by Claudia Felser’s group [6]). This is the reason in the literature involving such theoretical studies [7, 8, 16], people have always considered ratios.

14. The data availability statement says all data is included in the article, but the numerical data for the plots is not provided. Would the authors consider making this data available to facilitate further research?

The numerical data is in principle already provided because they precisely represent the values of the curves shown in the manuscript. However, if someone requests those in a table format as a data file, that will be provided. We do not have any university repository where I could have uploaded those data files in the first place.

15. How do the findings for this pseudospin-1 TSM compare with known results for the more common pseudospin-1/2 Weyl semimetals? Does the multifold nature lead to qualitatively new phenomena?

We have added the following lines:

“Our results show that a nonzero longitudinal conductivity in TSMs is caused by chiral anomaly, similar to the WSMs. The inclusion of OMM also leads to sign reversal of $\delta\sigma_{zz}$ depending on the relative strengths of internode and intranode scatterings, similar to the WSMs. The qualitative difference from the WSM-cases is that, since more than one band is participating in transport at each chiral node, we now have multiple knobs available to tune the characteristics of the linear response. These comprise various types of intraband and intranode scattering-amplitudes, viz., the set comprising $\{\beta_{s,\tilde{s}}^{\text{intra}}, \beta_{s,\tilde{s}}^{\text{inter}}\}$. Quantitatively, we find that a large ratio of internode-to-intranode scattering-strength is required (cf. Fig. 6) to get the sign-flips of $\delta\sigma_{zz}$, compared to WSMs [7].”

16. Are the predicted effects (e.g., the tunable sign of $\delta\sigma_{zz}$) strong enough to be observable in current experimental setups, and at what typical magnetic field strengths?

Yes, there are already some experiments by Claudia Felser’s group [6], where they have considered CoSi. They have measured negative magnetoresistance in planar-Hall set-ups with non-collinear electromagnetic fields, but fitted their data with a hand-waiving argument involving RTA.

LIST OF CHANGES

The newly added text in the revised manuscript has been shown in the color red.

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